

Data requirements for empirical estimation of comfort-zone boundaries

Active inference approach to driver comfort-zone boundaries

Chalmers University of Technology — WaymoActiveInference project

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1 Purpose and objectives

This document specifies exactly what data is needed to take the comfort-zone boundary method described in `notes/04_comfort_zone_method.md` from a construction to an empirical result. It is written so that it can be handed to a data owner, or used as a checklist when assessing whether an existing dataset is usable.

The objectives the data must support are, in order:

Primary objective. Calibrate the comfort-zone boundary level c against observed driver response onsets, and test whether a single c , fitted on one scenario type, predicts response onsets in held-out scenario types. This is the claim that distinguishes the method from conventional per-scenario indicator thresholds, and it is the reason more than one scenario type is required.

Secondary objective. Estimate the preference-function parameters that determine whose comfort zone is being described — desired speed, preferred inverse tau, assumed reaction time, and the assumed worst-case deceleration of the conflict partner — rather than adopting the values published by Schumann et al. (2026), which were tuned for their simulation study and not for any particular driver population.

Tertiary objective. Test whether "extra motives" in the sense of Näätänen and Summala move the boundary by the amount the method predicts. This requires either an experimental manipulation of time pressure, or observational data in which a proxy for hurry is recorded.

The authors of this document have not found any published work that estimates a comfort-zone boundary as a level set of a preference-derived scalar field, so there is no prior data specification to inherit. The requirements below are derived from the computation itself.

2 What the computation actually needs

The comfort-zone field is

$$\text{eps}(t) = \max_o \log p(o) - \log p(o(t))$$

where $p(o)$ is the driver's preference function over observations. Evaluating this along a recorded event requires only the observation vector at each time step. No model roll-out is performed, no particle filter is run, and no GPU is required. The function that consumes the data is `comfortzone.calibrate.deficit_along_trajectory`.

This matters for data procurement: the method needs *kinematics and geometry*, not driver state, not video, and not any signal that is expensive to obtain or ethically sensitive.

3 Required signals

Sampling rate at least 10 Hz; 20 to 50 Hz preferred. All signals must share one time base.

3.1 Ego vehicle

Signal	Symbol	Unit	Required	Used by	Notes
Time	t	s	yes	all	monotonic, uniform step preferred
Longitudinal position	x	m	yes	collision, safety	any fixed frame, see §4
Lateral position	y	m	yes	lateral	signed offset from lane center is acceptable and often better
Heading	ψ	rad	yes	collision severity	relative to lane direction
Speed	v	m/s	yes	speed, safety, tau	accuracy 0.1 m/s or better
Longitudinal acceleration	a_x	m/s ²	yes	control effort, safety	measured or differentiated from v

Signal	Symbol	Unit	Required	Used by	Notes
Road-wheel steering angle	delta	rad	preferred	control effort	steering wheel angle plus ratio is acceptable
Steering rate	omega	1/s	derived	control effort	differentiate delta if not recorded
Yaw rate	r	rad/s	fallback	control effort	only if no steering signal exists

3.2 Conflict partner

Signal	Symbol	Unit	Required	Used by	Notes
Longitudinal position	x_p	m	yes	collision, safety, tau	same frame as ego
Lateral position	y_p	m	yes	collision, lateral relevance	this is the signal most often missing or poor
Heading	psi_p	rad	yes	collision severity	cos(psi - psi_p) enters the severity factor
Speed	v_p	m/s	yes	safety, tau	
Acceleration	a_p	m/s^2	yes	safety	may be differentiated from v_p, see §7.2
Object class	—	—	yes	interpretation	car, truck, motorcycle, pedestrian, cyclist

If more than one road user is present, supply all of them that are plausibly relevant; the preference function takes a minimum over conflict partners.

3.3 Vehicle geometry

Quantity	Unit	Required	Notes
Ego length, width	m	yes	enters the collision box and the 1.15 L term
Partner length, width	m	yes	default values by class are acceptable if unknown, but flag them
Position reference point	—	yes	front bumper center, rear axle center, or center of gravity — see §4.3

3.4 Road geometry

Quantity	Unit	Required	Notes
Lane width	m	yes	the lateral preference is defined relative to it
Lateral offset of ego from lane center	m	yes	may be given directly instead of absolute y
Number of lanes, and travel direction of each	—	yes	same-direction and opposite-direction lanes are treated differently
Distance to road edge	m	preferred	needed for the off-road term
Curvature	1/m	optional	not used in the current implementation

3.5 Event annotations

Quantity	Unit	Required	Notes
Brake response onset	s	yes for the primary objective	this is the label the boundary level is fitted to
Steering response onset	s	preferred	needed for scenarios avoided by steering
Conflict onset	s	yes	when the partner's behavior created the conflict
Event outcome	—	preferred	crash, near-crash, conflict, or normal
Scenario type	—	yes	required for the generalization test

4 Coordinate frames, units, and sign conventions

Ambiguity here is the most common source of silent error, so the conventions are stated explicitly.

4.1 Frame

A right-handed frame with **x forward along the lane** and **y to the left**. Heading is positive counterclockwise. Either a fixed world frame or the ego frame is acceptable, provided both road users are expressed in the same one.

4.2 Relative quantities

The implementation uses

```
dx = x_partner - x_ego      positive when the partner is ahead
dy = y_partner - y_ego      positive when the partner is to the left
```

If a dataset provides range and azimuth instead, converting is straightforward, but the azimuth accuracy determines the quality of dy , which is often the limiting factor (§7.1).

4.3 Position reference point

State which point on the vehicle the coordinates refer to. The safety term contains a 1.15 ($l_f + l_r$) clearance allowance that assumes positions are vehicle *centers*. Supplying front-bumper positions without saying so shifts every boundary by roughly half a vehicle length, which is comparable to the effect being measured.

4.4 Units

SI throughout: meters, seconds, radians, m/s, m/s². If the source uses km/h, degrees, or feet, convert once at ingest and record that the conversion happened.

5 Duration and coverage per event

Each event should include:

- **At least 5 s before conflict onset.** This baseline is used to estimate the driver's normal following behavior, which is how the desired speed and preferred inverse tau are estimated for the secondary objective. Without it, those parameters must be assumed.
- **At least 10 s after conflict onset**, or until the conflict resolves, whichever is shorter.
- **No gaps in the partner track** during the conflict. If dropouts occur, flag them per sample rather than interpolating silently.

6 File format

Two files per dataset: one time series file per event, and one metadata table covering all events. Comma-separated text is sufficient; Parquet is preferable for large sets.

6.1 Time series file, one per event

File name <event_id>.csv. Column names exactly as below; extra columns are ignored.

```
t, ego_x, ego_y, ego_psi, ego_v, ego_ax, ego_delta,
oth_x, oth_y, oth_psi, oth_v, oth_ax, oth_class, valid
0.00, 0.00, 0.02, 0.001, 15.24, 0.05, 0.002, 26.90, 0.01, 0.000, 15.11, 0.00, car, 1
0.05, 0.76, 0.02, 0.001, 15.24, 0.03, 0.001, 27.66, 0.01, 0.000, 15.11, 0.00, car, 1
0.10, 1.52, 0.02, 0.001, 15.25, 0.02, 0.001, 28.41, 0.01, 0.000, 15.11, 0.00, car, 1
...
```

valid is 1 when the partner track is measured and 0 when it is interpolated or missing.

6.2 Metadata table, one row per event

```
event_id, dataset, scenario, driver_id, lane_width, n_lanes_same, n_lanes_opposite,
ego_len, ego_wid, oth_len, oth_wid, ref_point,
t_conflict_onset, t_brake_onset, t_steer_onset, outcome, notes
E0001, SIM_LTAP0D, ltap_0d, D07, 3.50, 1, 1, 4.62, 1.83, 4.40, 1.80, cog,
    6.20, 7.85, NaN, near_crash,
E0002, NDS_REAR, rear_end, D07, 3.65, 2, 0, 4.62, 1.83, 4.90, 1.90, front_bumper,
    12.40, 13.55, NaN, conflict, radar_lateral_position_poor
```

Use NaN for a response that did not occur. Do not use 0.

6.3 Optional per-driver table

If individual differences are of interest, a per-driver table with age, sex, annual mileage, and any experimental condition assignment.

7 Data quality: the failure modes that matter

These are the problems that have the largest effect on the estimated boundary, in order.

7.1 Lateral position of the conflict partner

Radar-based naturalistic driving data typically has good range and range rate but poor lateral position. The lateral preference term, the collision box, and the condition that restricts the safety term to same-lane partners all depend on dy . If lateral accuracy is worse than roughly 0.5 m, the same-lane and adjacent-lane cases cannot be separated reliably, and the method degrades to a purely longitudinal analysis. That is still usable for the primary objective in rear-end scenarios, but it rules out the cross-scenario generalization test.

7.2 Partner acceleration

The safety term uses $a_{\text{test}} = \min(a_{\text{partner}}, a_{0V_{\text{min}}})$, so a noisy a_{partner} propagates directly into the boundary. Differentiating speed at 10 Hz with a low-pass filter is normally adequate. Verify by checking that the differentiated acceleration during steady following is centered on zero with a standard deviation below about 0.3 m/s^2 . Report the filter used.

7.3 Time synchronization

Vehicle CAN signals and external perception are often logged by different systems. An offset of 100 ms shifts the estimated response time by the same amount, which is a large fraction of the effect being studied. Confirm synchronization, and state the residual uncertainty.

7.4 Response onset definition

Different datasets define onsets differently, and mixing definitions across scenario types would confound the generalization test. Use one definition throughout. Two acceptable options:

- 1 **Piecewise-linear fit to the speed trace**, taking the instant at which the first constant segment turns into a falling one. This is the definition used by Markkula et al. and adopted by Schumann et al., so results are comparable with the published values.
- 2 **Threshold crossing**, typically -1 m/s^2 longitudinal acceleration for braking and 5 degrees of steering wheel angle for steering.

Whichever is used, record it in the metadata and apply it identically to every scenario.

7.5 Missing lane geometry

Naturalistic data often lacks lane width and lane assignment. A constant nominal lane width can be substituted, but then the lateral term becomes approximate and any boundary that depends on lateral position should be treated as indicative rather than estimated.

8 Sample size

These are working estimates based on the shape of the fitting problem, not on a formal power analysis; a power analysis should be run once pilot variance is known.

Purpose	Events needed	Notes
Fit one boundary level c by onset matching	30 to 50 per scenario type	the fit criterion is flat near its optimum, so precision improves slowly beyond this
Cross-scenario generalization test	at least 2 scenario types, preferably 3, with 30+ each	this is the primary objective
Distributional calibration without onset labels	200+ events	uses an acceptance quantile instead of onsets
Individual differences in c	10+ events per driver, 20+ drivers	
Extra-motive manipulation	20+ events per condition per driver	within-subject design strongly preferred

9 Candidate datasets

Suitability below refers to the primary objective unless stated otherwise. Access status should be confirmed; the assessment here is based on published descriptions rather than on inspection of the data.

Dataset	Kinematics	Partner lateral	Lane geometry	Onset labels	Suitability
Chalmers LTAP/OD comfort-zone study	good	good	known	likely	highest — boundaries already quantified conventionally, so it directly tests whether the level set reproduces an existing result

Dataset	Kinematics	Partner lateral	Lane geometry	Onset labels	Suitability
Chalmers pedestrian-overtaking field test and UDRIVE analysis	good	good	known	may need re-extraction	high — a different geometry, which is what the generalization test requires
Driving simulator, new collection	perfect	perfect	perfect	perfect	high — the only route to the tertiary objective, since time pressure can be manipulated
SHRP2 naturalistic driving study	good	moderate to poor	partial	yes, for near-crashes	high for rear-end; the dataset underlying the deceleration comparison in Schumann et al.
UDRIVE	good	moderate	partial	partial	high, and already used in Chalmers comfort-zone work
euroFOT	moderate	poor	limited	limited	moderate
highD, exiD, inD, round (levelXdata, drone)	excellent	excellent	excellent	none	good for distributional calibration only; no driver state and no onsets
pNEUMA (Athens, drone)	good	good	moderate	none	as above, urban
Waymo Open Motion Dataset	excellent	excellent	excellent	none	good for distributional calibration; also the closest match to the data the original model was built around
nuScenes, Argoverse 2	good	good	good	none	short sequences limit pre-conflict baseline
ANNEXT	unknown	unknown	unknown	unknown	referenced by Schumann et al. as an African naturalistic dataset; the present author has not established its availability

9.1 Recommended combination

If two datasets can be obtained, the combination with the most scientific value is:

- 1 **Chalmers LTAP/OD** — fit c here, where a conventional comfort-zone boundary already exists for comparison.
- 2 **Chalmers pedestrian overtaking** — apply the same c without refitting, and test whether it predicts onsets in a geometry the fit never saw.

A simulator study is the natural third component, because it is the only way to manipulate extra motives rather than infer them.

10 What is not needed

Stating this explicitly reduces the burden on the data owner:

- **Video**, except where it is needed to produce or verify the onset labels
- **Driver physiological data** — heart rate, skin conductance, and similar
- **Eye tracking**, for the comfort-zone work as specified. It becomes relevant only if the epistemic-value component of the model is brought in later, for example to model occlusion handling or off-road glances
- **High-definition maps** beyond lane width, lane count, and travel direction
- **Any personally identifying information**

11 Minimum viable dataset

If only one thing can be obtained, the smallest set that supports a meaningful result is:

- approximately 40 rear-end conflict events,
- kinematics at 20 Hz for ego and lead vehicle, including both accelerations,
- lane width and vehicle dimensions,
- brake onset labeled by a single stated definition,
- at least 5 s of pre-conflict baseline per event.

That supports the secondary objective in full and the primary objective within one scenario type. It does not support the generalization test, which is the claim most worth testing.

12 Limitations of this specification

Volunteered rather than left for a reviewer to raise:

- The specification assumes the preference function of Schumann et al. (2026), including the lane-structured lateral term and the counterfactual braking margin. A different preference function would change which signals matter, though the kinematic core would not change.
- The lateral preference is defined per scenario in the original work rather than derived from a map. Applying it to a new scenario type requires a modeling decision that this specification cannot remove.
- Response onset is treated as the behavioral marker of comfort-zone exceedance. That is the assumption the primary objective depends on, and it is inherited from the surprise-based response-timing literature rather than established independently.
- No formal power analysis has been performed; the sample sizes in §8 are working estimates.

13 References and verification status

Each reference is marked with how it was obtained, following the project convention.

Reference	Status
Schumann, J. F., Engström, J., Johnson, L., O'Kelly, M., Messias, J., Kober, J. and Zgonnikov, A. (2026). Active inference as a model of collision avoidance behavior in human drivers. <i>Nature Communications</i> 17:5009. doi:10.1038/s41467-026-73345-0	read from source PDF , including the Supplementary Information
Dinparastdjadid, A., Supeene, I. and Engström, J. (2023). Measuring surprise in the wild. arXiv:2305.07733	read from source PDF
Engström, J., Wei, R., McDonald, A. D., Garcia, A., O'Kelly, M. and Johnson, L. (2024). Resolving uncertainty on the fly. <i>Frontiers in Neurorobotics</i> 18:1341750	read from source PDF
Modirshanechi, A., Brea, J. and Gerstner, W. (2022). A taxonomy of surprise definitions. <i>Journal of Mathematical Psychology</i> 110:102712	read from source PDF
Engström, J., Bärgman, J., Nilsson, D., Seppelt, B., Markkula, G., Piccinini, G. B. and Victor, T. (2018). Great expectations: a predictive processing account of automobile driving. <i>Theoretical Issues in Ergonomics Science</i> 19(2):156–194	verified by search — title, journal, volume and pages confirmed; full text not read
Quantifying drivers' comfort-zone and dread-zone boundaries in left turn across path / opposite direction (LTAP/OD) scenarios. <i>Transportation Research Part F</i> (2015)	verified by search — title and journal confirmed via ScienceDirect and the Chalmers research portal; author list not verified , full text not read
How do drivers overtake pedestrians? Evidence from field test and naturalistic driving data. <i>Accident Analysis and Prevention</i> (2019)	verified by search — title and journal confirmed; author list not verified
Markkula, G. et al., brake response extraction by piecewise-linear fit	unverified — the method is described in Schumann et al. (2026) Methods, which cites it; the original was not consulted
Näätänen, R. and Summala, H. (1974, 1976), extra motives and zero-risk theory	unverified — referred to in secondary sources encountered during this work; originals not consulted
SHRP2, UDRIVE, euroFOT, highD, exiD, pNEUMA, Waymo Open Motion Dataset, nuScenes, Argoverse 2	unverified for this purpose — the assessments in §9 are based on general knowledge of these datasets and should be confirmed against their documentation before procurement

Gaps deliberately left visible rather than filled with plausible detail: the author lists for the two Chalmers comfort-zone papers, and the availability of ANNEXT.